Traffic Density Estimation Using CSRNet

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***Abstract*—Traffic congestion is a common problem in modern cities and directly impacts travel time, fuel consumption, and pollution levels. Accurate traffic density estimation can help in better traffic monitoring and management. In this project, we use CSRNet, a deep learning-based density estimation model, to estimate traffic density from surveillance images. Instead of detecting individual vehicles, CSRNet generates a density map and estimates the vehicle count by integrating the predicted map. This approach is effective in handling heavy traffic, occlusion, and varying lighting conditions.**

***Index Terms*—Traffic density estimation, CSRNet, Deep learning, Density maps, Intelligent transportation systems**

I. Introduction

With the rapid increase in vehicle usage, traffic congestion has become a serious issue in urban areas. Efficient traffic monitoring systems are required to manage congestion and improve road safety.

Traditional traffic monitoring techniques rely on hardware sensors or manual counting, which are expensive and not always scalable. Computer vision-based approaches offer a more flexible solution by using existing surveillance cameras.

In this project, we focus on density-based traffic estimation using CSRNet. Unlike object detection methods that detect each vehicle separately, density estimation models predict a continuous density map that represents vehicle concentration in the scene.

II. Motivation

Detecting individual vehicles in crowded traffic scenes is challenging due to occlusion and overlapping vehicles. Density estimation methods provide a more stable and scalable solution. CSRNet has shown strong performance in crowd counting tasks, and this project adapts it for traffic density estimation.

III. Proposed Methodology

The proposed workflow consists of the following steps:

1) Collecting traffic images from publicly available datasets.
2) Preprocessing images through resizing and normalization.
3) Generating ground truth density maps.
4) Training CSRNet to predict density maps.
5) Estimating total vehicle count from the predicted density map.

The vehicle count is computed as:

$$Count = \sum_{x} D(x) \tag{1}$$

where $D(x)$ represents the predicted density value at pixel location x.

IV. CSRNet Overview

CSRNet consists of:

- A front-end based on VGG-16 for feature extraction.
- A back-end with dilated convolution layers to increase receptive field while preserving spatial resolution.

Dilated convolutions help capture contextual information without reducing image resolution, which is important for dense traffic scenes.

V. Tools and Technologies

- Python
- PyTorch
- OpenCV
- NumPy and Matplotlib

VI. Expected Outcome

The expected outcome is a trained CSRNet model capable of generating accurate density maps and estimating vehicle count from traffic images. The system should perform reliably even in crowded and complex traffic conditions.

VII. Future Scope

Future improvements may include real-time implementation, integration with smart traffic signal systems, and lane-wise traffic analysis.

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